Ex:- $V = C[a,b]$

$$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt$$

I1 $$\langle \alpha f_1 + \beta f_2, g \rangle = \int_a^b \{\alpha f_1(t) + \beta f_2(t)\}\, g(t)\, dt$$

$$= \int_a^b \alpha f_1(t)\, g(t)\, dt + \int_a^b \beta f_2(t)\, g(t)\, dt \quad [\because \text{integral operator is linear}]$$

$$= \alpha \int_a^b f_1(t)\, g(t)\, dt + \beta \int_a^b f_2(t)\, g(t)\, dt$$

$$= \alpha \langle f_1, g \rangle + \beta \langle f_2, g \rangle$$

I2 $$\langle f, g \rangle = \int_a^b f(t)\, g(t)\, dt$$

$$= \int_a^b g(t)\, f(t)\, dt$$

$$= \langle g, f \rangle$$

I3 $$\langle f, f \rangle = \int_a^b f(t)\, f(t)\, dt$$

$$= \int_a^b (f(t))^2\, dt \geq 0$$

If $\langle f, f \rangle = 0$

$\Leftrightarrow \int_a^b (f(t))^2\, dt = 0$

$\Leftrightarrow f(t) = 0$

Pkalika Maths